

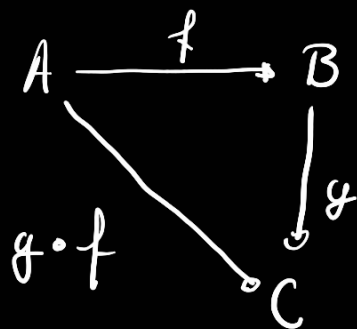
Cors 2 Algebră

○ funcția este

a) respectivă p.t. $x_1, x_2 \in A, x_1 \neq x_2$ implică $f(x_1) \neq f(x_2)$

... ? nu știu ce să scriu

Ex. 1.3.15.



$$(a) \ x_1, x_2 \in A \text{ c. } (g \circ f)(x_1) = (g \circ f)(x_2) \Rightarrow g(f(x_1)) = g(f(x_2)) \xrightarrow{g} \text{inj} \\ \Rightarrow f(x_1) = f(x_2) \text{ (in } B) \xrightarrow{f} \text{inj} x_1 = x_2. \text{ Deci } g \circ f \text{ este inj}$$

$$(b) \text{ Fie } z \in C \xrightarrow{g} \text{inj}$$

(c) este (a) & (b)

$$(d) \text{ Fie } x_1, x_2 \in A \text{ c. } f(x_1) = f(x_2) \xrightarrow[\text{funcția}]{g} g(f(x_1)) = g(f(x_2))$$

$$(g \circ f)(x_1) = (g \circ f)(x_2) \xrightarrow[\text{inj}]{g \circ f} \begin{matrix} \uparrow \\ B \end{matrix} x_1 = x_2 \text{ deci } f \text{ este inj}$$

(e) Fie $z \in C \xrightarrow[\text{surj.}]{g \circ f} \exists x \in A : (g \circ f)(x) = z$

Nel $y = f(x) \in B$

$$g(y) = g(f(x)) = (g \circ f)(x) = z$$

Obs: $A \xrightarrow{f} B$ $g \circ f = 1_A \Leftrightarrow$
 $\forall x \in A (g \circ f)(x) = 1_A(x) \Leftrightarrow$
 $\forall x \in A g(f(x)) = x$

1.3.16

(i) $\Rightarrow$ (ii) Fie $y \in B$

Deci $y \in f(A) \Rightarrow \exists x \in A : f(x) = y$

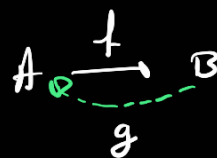
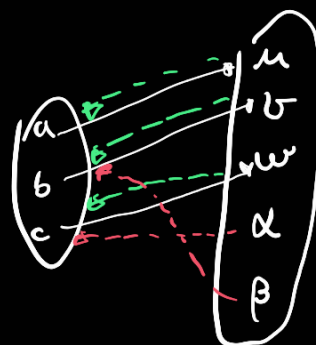
f -inj $\Rightarrow \exists! x \in A : f(x) = y$

in acest caz def. $g(y) = x$

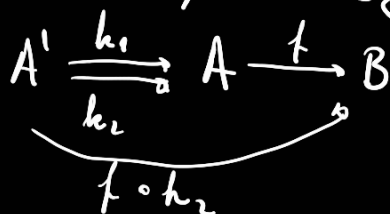
Deci $y \notin f(A)$ p.c. $A \neq \emptyset \exists a_0 \in A$ si punem $g(f) = a_0$

Atunci $g(f(x)) = x$ deci $g \circ f = 1_A$

Exemplu



(ii) $\Rightarrow$ (iii) Sfirm ca $\exists g : B \rightarrow A$ cu $g \circ f = 1_A$



$$f \circ h_1 = f \circ h_2 \Rightarrow g \circ (f \circ h_1) = g \circ (f \circ h_2) \Rightarrow (g \circ f) \circ h_1 = (g \circ f) \circ h_2$$

$$1_A \circ h_1 = 1_A \circ h_2 \Rightarrow h_1 = h_2$$

(iii) $\Rightarrow$ (i) Fie $x_1, x_2 \in A: f(x_1) = f(x_2) \in B$

Fie $A' = \{0\}$ și $h_1, h_2: A' \rightarrow A, h_1(0) = x_1, h_2(0) = x_2$

$$f \circ h_1, f \circ h_2: A' \rightarrow B$$

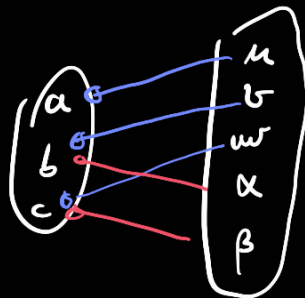
$$\left. \begin{aligned} (f \circ h_1)(0) &= f(h_1(0)) = f(x_1) \\ (f \circ h_2)(0) &= f(h_2(0)) = f(x_2) \end{aligned} \right\} \Rightarrow f \circ h_1 = f \circ h_2 \stackrel{(iii)}{\Rightarrow} h_1 = h_2$$

$x_1 = h_1(0) = h_2(0) = x_2$ Deci f este inj.

1.3.17

(i) $\Rightarrow$ (ii) g surj. $\Rightarrow \forall x \in A: g^{-1}(\{x\}) = \{y \in B: g(y) = x\} \neq \emptyset$

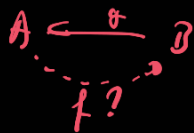
Exemplu



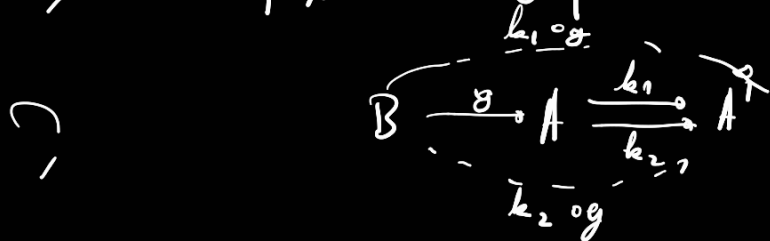
Alleg $y_x \in g^{-1}(\{x\})$ și definim $f(x) = y_x$

$$\forall x \in A: g(f(x)) = g(y_x) = x \text{ pt. c\u0102}$$

$$y_x \in g^{-1}(\{x\}). \text{ Deci } g \circ f = 1_A$$



(ii) $\Rightarrow$ (iii) Stim c\u0102 $\exists f: A \rightarrow B$ cu $g \circ f = 1_A$



(iii) $\Rightarrow$ (i)

Presupunem c\u0102 g nu este surj. $\Rightarrow \exists a \in A: g(y) \neq a \quad \forall y \in B$

Fie $A' = \{0, 1\}$ și $h_1, h_2: A \rightarrow A'$

$$h_1(x) = 0, \forall x \in A \quad h_2(x) = \begin{cases} 0, & x \neq a \\ 1, & x = a \end{cases}$$

Atunci $h_1 \neq h_2$

$$\left. \begin{aligned} \text{Dar } h_1 \circ g, h_2 \circ g: B &\rightarrow A' \\ (h_1 \circ g)(y) &= h_1(g(y)) = 0 \\ (h_2 \circ g)(y) &= h_2(g(y)) = 0 \end{aligned} \right\} \Rightarrow h_1 \circ g = h_2 \circ g$$

+
a

Deci g nu e simplificabil

CARDINALUL UNEI MULTIMI

1.3.21 TEM 7 !!

$y \in \mathcal{P}(A)$ same thing as $x \subseteq A$

